TESTING RECORD

OT Project Description Builder: Testing Record

One wave of live testing across five OT scenarios in April 2026: NDC prototype, SOW conversion, BAA consortium with competition commitment, and research OT. 15 universal patches shipped, zero regressions on post-patch runs.

Skill
ot-project-description-builder
github.com/1102tools/federal-contracting-skills

Date
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1102tools.com

The bottom line

One testing wave in April 2026 across five end-to-end scenarios on Claude Opus 4.7 shipped 15 patches to the OT Project Description Builder. All 15 patches validated on regression across two post-patch runs that exercised untested territory (software prototype, consortium, path D competition commitment, research OT, traditional prime). No regressions, no new universal gaps.

The skill reliably produces 10 USC 4021 and 4022 compliant project descriptions across prototype, research, and production-follow-on paths, with milestone-based payment structures, TRL-mapped phase progression, data rights handling, cost-sharing path selection, and a separate chat-only milestone handoff table for the OT Cost Analysis.

Scenarios tested

#	Scenario	Authority	Performer path	Workflow
1	Ultra-lazy drone prototype	10 USC 4021	NDC	A (full build)
2	Tethered surveillance drone, richly specified	10 USC 4021	NDC, 4022(d)(1)(A)	A
3	FAR-flavored SOW conversion with CLIN/cost contamination	10 USC 4021	Traditional, 4022(d)(1)(C)	B (conversion)
4	NSTXL-brokered BAA white paper, software prototype	10 USC 4021 + 4022(f)	Traditional with competition commitment, 4022(d)(1)(D)	B
5	Cold-weather battery chemistry research	10 USC 4021 (research)	Traditional, no cost-share required	A

All five runs produced contract-file-ready .docx outputs, emitted the Milestone Handoff Table as chat output only, and stopped at the Phase 2 Invocation Gate for user "proceed" before document generation.

What the skill got right on every run

- Four-question Acquisition Context Intake up front (OT type, performer type, TRL entry/exit, consortium/direct)
- Phase 2 Invocation Gate held. No self-approval observed on any run after the gate patch landed.
- Milestone Handoff Table presented as chat-only markdown, never embedded in the .docx or saved as a separate file
- Section 12 Cost-Sharing conditionally omitted for NDC, SB, competition-commitment paths, and for 10 USC 4021 research authority (where 4022(d) is statutorily inapplicable)
- No FAR clause references leaked into the body. The one permitted exception (DFARS 252.227-7013/7014 cited as data rights taxonomy framework under OT tailoring) handled correctly.
- No cost figures, CLINs, FTE counts, or SOC codes in any document body
- Traditional SOW-to-OT conversion correctly stripped CLIN structure, FAR 52 clauses, QASP/AQL language, and dollar amounts while preserving scope intent

Patches shipped in this wave

Patches were applied in two sub-groups. Group 1 was cross-shipped from the SOW/PWS Builder Wave 2 patch set. Group 2 was derived from three live OT runs and validated against two subsequent runs covering untested territory.

Group 1: Cross-shipped from SOW/PWS Wave 2 (5 patches)

Patch	Trigger
Phase 2 Invocation Gate with "DO NOT self-approve" language	Universal pattern in document-generating skills
Anti-redundancy rule (do not re-ask questions the user's prompt already answers)	Universal pattern
AskUserQuestion guidance for Phase 1 intake batching	Matches claude.ai web chat behavior
UNCONDITIONAL RULE requiring Milestone Handoff Table emission	Mirrors SOW/PWS unconditional handoff pattern
Section ordering prescriptive with explicit renumbering rule for omitted conditional sections	Prevents drift across runs

Group 2: OT-specific universal gaps surfaced in Tests 1-3 (10 patches)

Patch	Section affected	Trigger
Corrected NDC definition to CAS full-coverage test under 10 USC 3014, removed the incorrect "$500K+ in prior year" threshold	Intake Q2	Test 2 flagged the factual error; verified against statute
Conditional docx generator path (use `/mnt/skills/public/docx/SKILL.md` on claude.ai sandbox, fall back to python-docx on Claude Code and local installs)	Phase 2 opening	Test 2 flagged the sandbox-only path; all three tests had workarounds
Split canned handoff closing message into two variants (cost-share and no-cost-share)	Phase 3 handoff	The single version incorrectly promised "apply cost sharing" on NDC, SB, and competition paths
Required Document Review Checklist emission in chat (with per-item pass or attention status) before the Milestone Handoff Table	Phase 3	All three initial runs ran the checklist mentally and moved on
Go/no-go gate carry-through rule (phase-boundary milestones must carry Block 2 Q7 criteria into both Section 4 and Section 5)	Phase 1 milestone derivation	Test 1 flagged; criteria collected but not surfaced in the document
Defined "Est. Duration" column semantics (calendar months from prior milestone)	Phase 3 handoff table	Ambiguous durations produce 3x+ cost-analysis labor loading deltas
Cost data stripping rule for Workflow B (strip source dollar figures from body, pass to handoff as informational only)	Phase 0 Document Intake	Test 3 needed this; skill was silent

Patch	Section affected	Trigger
Cost-share arithmetic disambiguation for Workflow B + 10 USC 4022(d)(1)(C)	Phase 0 Document Intake	Whether a stated total is government share or total agreement value is ambiguous in most SOWs and produces materially different government obligations
Performance placeholder closeout mechanism for [TBD] thresholds in Section 3 Objectives, with default bilateral-modification disposition at Phase 1 CDR	Phase 2 Section 14	Multiple runs invented disposition mechanisms; standardized here
Body cross-references must use section titles rather than section numbers	Phase 2 Section Structure	Prevents broken refs when conditional sections are omitted and subsequent sections renumber

Regression validation (Tests 4 and 5)

Tests 4 and 5 exercised untested territory after all 15 patches landed. They were designed specifically to stress parts of the skill that Tests 1-3 didn't touch.

- **Test 4 (BAA conversion, consortium, path D, software, late TRL 5-7):** Every Group 2 patch that could fire did fire correctly. The conditional handoff message used the no-cost-share variant. The Document Review Checklist emitted as 11 items in chat. Cross-reference patch held (no numbered refs in body). Source-doc cost stripping wrote "None. White paper contained no cost figures" to the handoff, demonstrating the rule holds even when there is nothing to strip.
- **Test 5 (Research OT, traditional prime, 30 months):** Skill correctly identified that 10 USC 4022(d) cost-sharing paths do not apply to 10 USC 4021 research authority and omitted Section 12. Cost-share arithmetic question did not fire because the statutory framework makes it inapplicable (the correct behavior, not a miss). Go/no-go criteria carried through as placeholder thresholds with stated disposition.

No regressions observed. No new universal gaps surfaced.

What was not tested

- **Workflow C (scope reduction)** for OT project descriptions. One test was drafted but not run in this wave. The skill defines Workflow C but its universal patches have not been regression-tested against a real scope-reduction conversation.
- **Production follow-on OT as primary workflow.** Tests 2 and 4 included 10 USC 4022(f) provisions as an option, but neither run structured the agreement as a production follow-on.
- **Consortium-specific templates.** Test 4 used NSTXL as the broker; DIU, AFWERX, NavalX, SOSSEC, and MTEC consortium variations have not been exercised.
- **Small business performer path (10 USC 4022(d)(1)(B)).** Tests covered NDC (path A), traditional with competition commitment (path D), traditional with cost-share (path C), and research (no 4022(d) applicability). Small-business significant participation was not tested.
- **Multi-performer consortia.** All tests used single performers or single primes with no subs.
- **Deep classified prototype contexts.** No SAP, SAR, or ICD 705 variants were exercised.
- **Hardware prototype beyond drone and ground vehicle.** Tests 1, 2, and 3 all touched airborne or ground-mobility prototypes. Directed energy, hypersonics, biotech, cyber tooling, and AI/ML prototypes beyond the Test 4 software case have not been exercised.

Users in these contexts should expect to validate outputs more carefully and may encounter edge cases this wave did not surface.

Testing methodology

Five live runs on Claude Opus 4.7 via claude.ai web chat and Claude Code, the same environments the skill's end users run in.

Evaluator grading was done by a separate Claude Opus 4.7 instance with access only to the skill text and the worker's final output transcript. Grading focused on three questions per run:

1. Did each patch under test fire correctly?
2. Did the output surface any new structural gap not covered by existing patches?
3. If a gap was observed, is it a universal pattern across the skill's full usage profile, or a one-off of model judgment on a specific input?

Only gaps classified as universal structural patterns produced new patches. Drone-specific feedback (unit count defaults, airworthiness prompting), DoD-specific feedback (CMMC, ITAR), and over-engineering suggestions (sentinel markers, trigger-phrase hardening) were deliberately skipped to prevent bloat.

Skill line count: 408 before the wave, 440 after all 15 patches (+32 lines net on a skill that runs hundreds to thousands of times).

Patches: shipped in this wave

Skill version lines: 408 before patches, 440 after. Ceiling remains 1,000.

All 15 patches are live in the current SKILL.md.

> **Testing Methodology**
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> Evaluator: James Jenrette (1102tools) and Claude Code Opus 4.7 (1M context window, max effort mode, Claude Max 20x subscription).
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> Worker model tested: Claude Opus 4.7 on claude.ai web chat and Claude Code, the same environments the skill's end users run in.
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> Wave: 5 runs, 15 patches shipped, 2 post-patch regression runs with zero new gaps surfaced.
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> Date: April 2026.
>
> Skill: ot-project-description-builder. Source: github.com/1102tools/federal-contracting-skills. License: MIT.